



PROBABILITY



Name: _____ Date: _____

★ LEVEL 4: EXPERIMENTAL PROBABILITY ★

1 COIN TOSS RESULTS.

A coin was tossed 12 times.
Here are the results.



Result	Number of times
Heads	7
Tails	5

- a) Which happened more often? _____
- b) What is the experimental probability of heads? _____
- c) What is the experimental probability of tails? _____

Remember: total tosses = 12

2 SPINNER TEST.

A spinner with 3 colours was spun 20 times.
Here are the results.



Colour	Number of times
Red	9
Blue	7
Yellow	4

- a) Complete the bar chart using the results above.



- b) Which colour occurred most? _____
- c) Which colour occurred least? _____
- d) Experimental probability of red = _____
- e) Experimental probability of blue = _____
- f) Experimental probability of yellow = _____

Remember: total spins = 20

3 BAG EXPERIMENT.

A bag of counters was picked from 15 times. Here are the results.



Shape	Number of times
Star	6
Heart	5
Circle	4

- a) Which shape seems most likely to be picked? _____
- b) Which shape seems least likely to be picked? _____
- c) Experimental probability of star = _____
- d) Experimental probability of heart = _____
- e) Experimental probability of circle = _____

Remember: total picks = 15

4 COMPARE THEORETICAL AND EXPERIMENTAL.



For a fair coin, the theoretical probability of heads is $\frac{1}{2}$ (one half).

In Activity 1, the coin was tossed 12 times and there were 7 heads.

- a) Is the experimental probability of heads ($\frac{7}{12}$) exactly the same as $\frac{1}{2}$? ☐ Yes ☐ No
- b) Is $\frac{7}{12}$ just close to $\frac{1}{2}$? ☐ Yes ☐ No
- c) Challenge! Explain your answer.



Great job! You are becoming a probability superstar!